

# Ultimatum Game Classroom Run Kit

TOK Cognitive Science Activity Bank • Human Sciences • 25-35 min

|            |                |      |                |
|------------|----------------|------|----------------|
| Category   | Human Sciences | Time | 25-35 min      |
| Difficulty | Medium         | ID   | ultimatum-game |

Big idea: Humans do not always maximise narrow utility.

## One-Minute Teacher Brief

Students test whether human decisions always match narrow economic rationality.

## Student Prompt

Inspect Ultimatum Game Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

## Concrete Stimulus Packet

|                               |                                                                                                                                                                                                                          |                                                                                                              |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Ultimatum Game Stimulus A     | Student-facing stimulus 1: Students split ten tokens and record accept/reject decisions. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.           | Students inspect this exact card first and commit to a claim before hearing anyone else.                     |
| Ultimatum Game Stimulus B     | Student-facing stimulus 2: Offer cards showing fair, unfair, strategic, and emotional proposals. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result.   | Use this for comparison, a second round, or a partner challenge.                                             |
| Ultimatum Game Reveal / Twist | Student-facing stimulus 3: A graph comparing rational-choice prediction with actual class choices. Predict how people will respond, then identify the wording, context, incentive, or role that could change the result. | Teacher reveal: connect the changed response to the big idea — Humans do not always maximise narrow utility. |

## Actual Lesson Questions

| # | Question for students                                                           | Teacher key / cue                                                                                             |
|---|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1 | After inspecting Ultimatum Game Stimulus A, what is your first knowledge claim? | Teacher cue: look for a clear claim, not just a description.                                                  |
| 2 | Which exact detail from Ultimatum Game Stimulus A did you use as evidence?      | Teacher cue: students should name visible words, data, source features, method features, or contextual clues. |
| 3 | What did you infer beyond the evidence?                                         | Teacher cue: this is where assumptions, framing, models, or perspective usually appear.                       |
| 4 | How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?                        | Teacher cue: confidence is data for the debrief, not a grade.                                                 |
| 5 | What missing information would most improve the knowledge claim?                | Teacher cue: push for method, source, context, comparison, or definitions.                                    |
| 6 | Can human behaviour be predicted scientifically?                                | Teacher cue: students should move from the classroom example to a general TOK issue.                          |

## Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Human Sciences.

## Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how humans do not always maximise narrow utility.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

## Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: Can human behaviour be predicted scientifically?

## Setup Checklist

- Prepare a starter stimulus using: Students split ten tokens and record accept/reject decisions.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

## Example Stimuli or Materials

- Students split ten tokens and record accept/reject decisions.
- Offer cards showing fair, unfair, strategic, and emotional proposals.
- A graph comparing rational-choice prediction with actual class choices.

## Resources To Use

- Resource pack: <resources/ultimatum-game/index.html>
- Card deck CSV: <resources/ultimatum-game/cards.csv>
- Visual prompt: <resources/ultimatum-game/visual-prompt.svg>
- Printable student worksheet with observation, inference, evidence, and confidence columns.
- Teacher notes with setup checklist, sample facilitation language, misconceptions, and assessment look-fors.
- Classroom run kit with prompt cards, example stimuli, and debrief moves.
- PowerPoint deck with a student-facing sequence for running the activity.

## Run Sequence

1. Pair students. Player A divides 10 tokens between A and B.
2. Player B accepts or rejects. If rejected, both get nothing.
3. Collect anonymous results.
4. Discuss why people reject unfair offers.

## **Debrief Moves**

- Knowledge question: Can human behaviour be predicted scientifically?
- Are humans rational knowers?
- How do fairness, emotion and culture affect decision-making?

## **Teacher Quick Script**

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

## **Exit Ticket**

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...